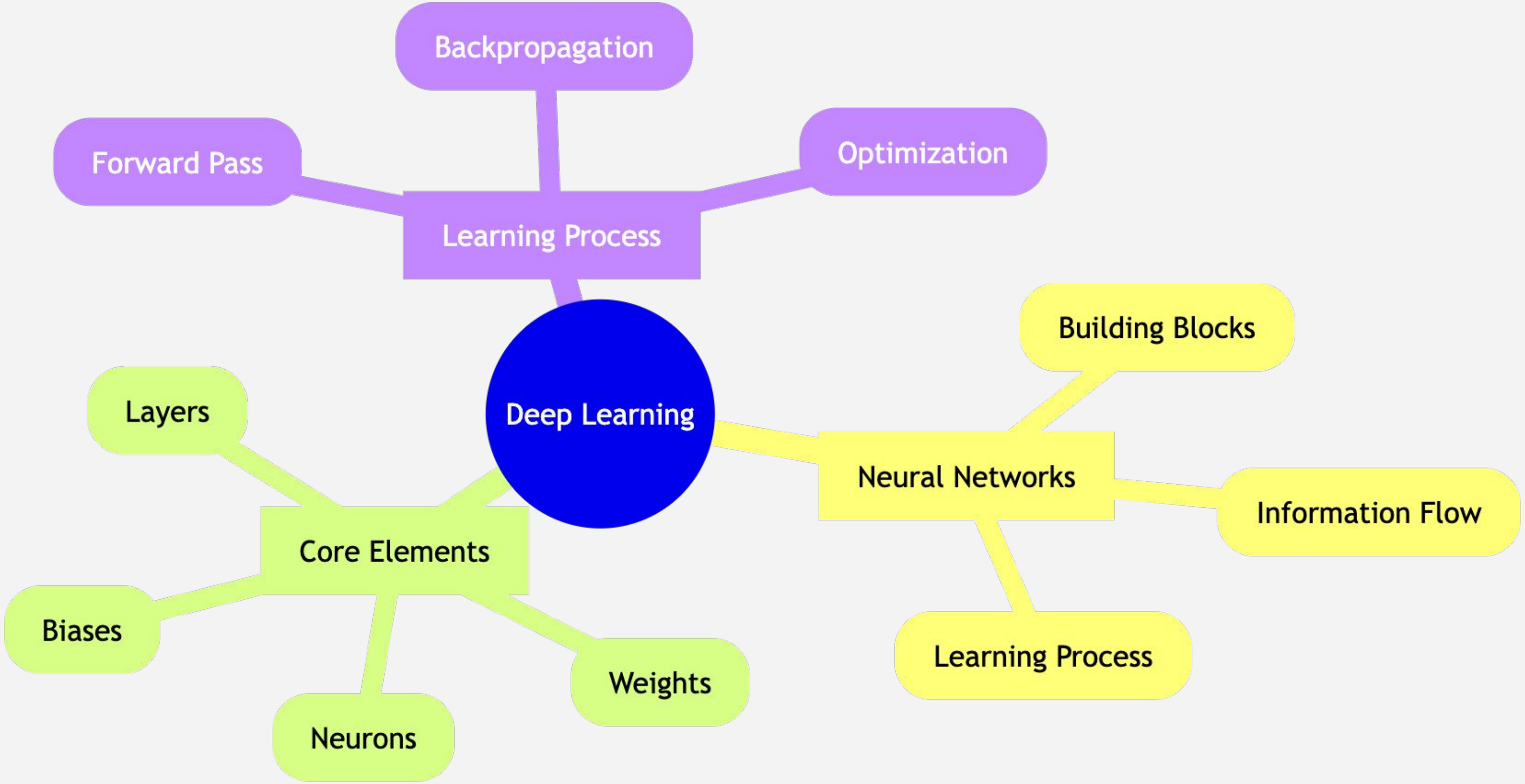
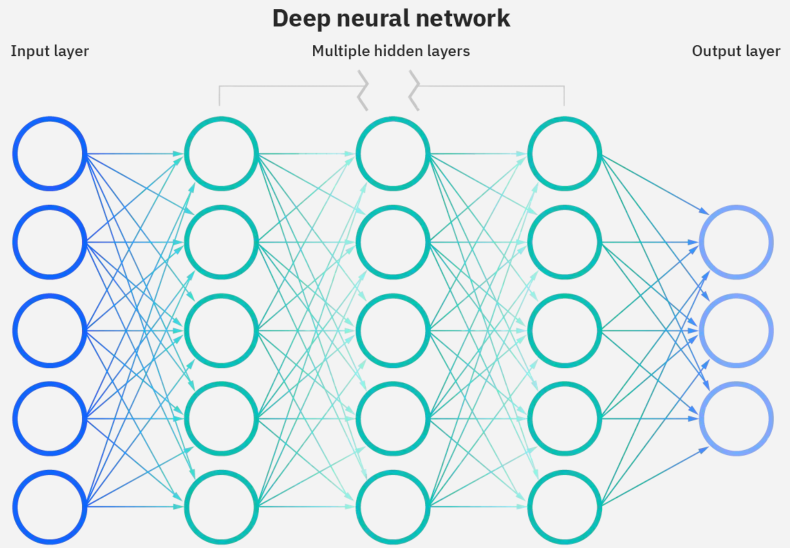
Deep Learning Key Concepts



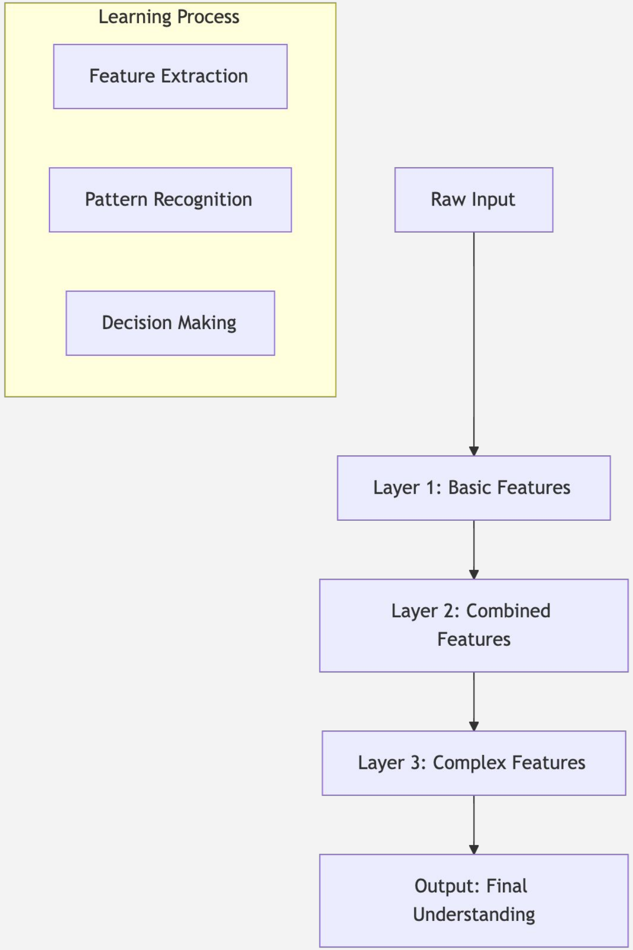
Deep Learning Overview

Neural networks with multiple layers learn from data by:

- Automatically discovering features
- Learning hierarchical representations
- Transforming raw input into desired output

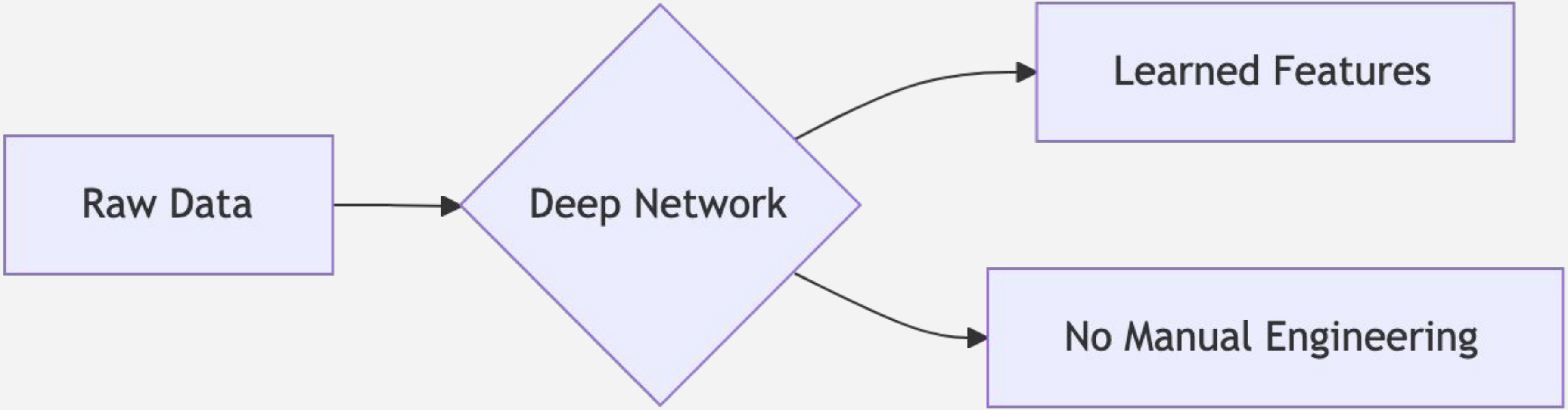


Deep Learning - Breakdown



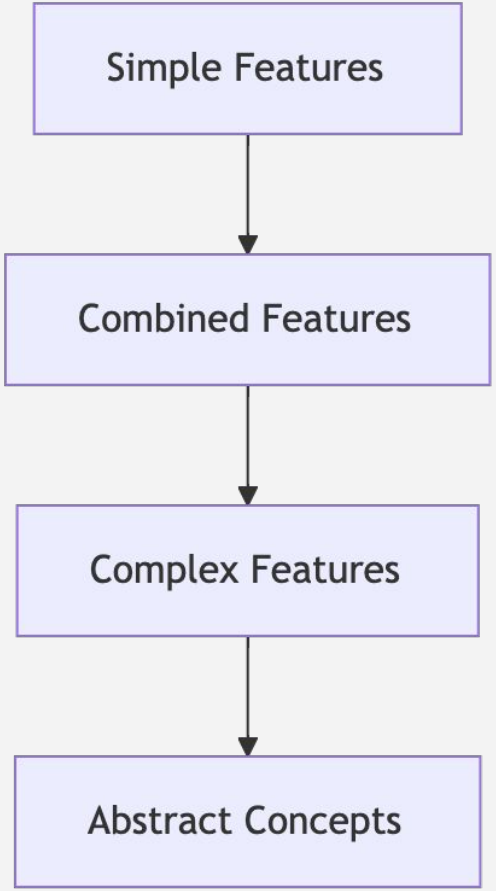
Deep Learning - Key aspects

1. Automatic Feature Learning



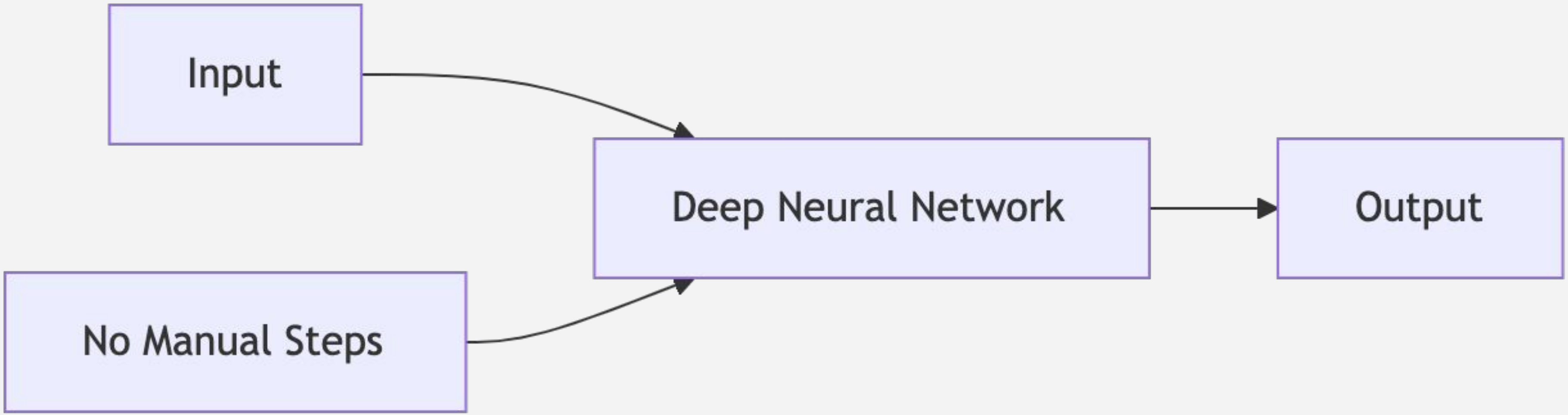
Deep Learning - Key aspects

2. Hierarchical Structure



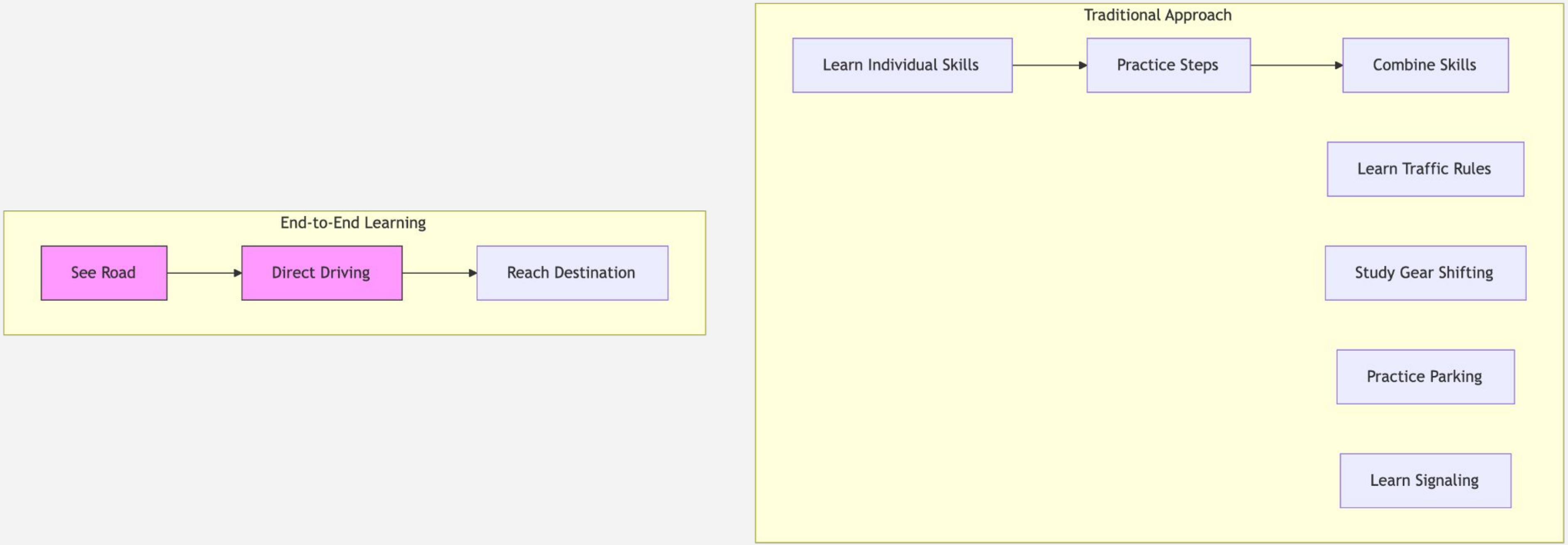
Deep Learning - Key aspects

3. End-to-End Learning



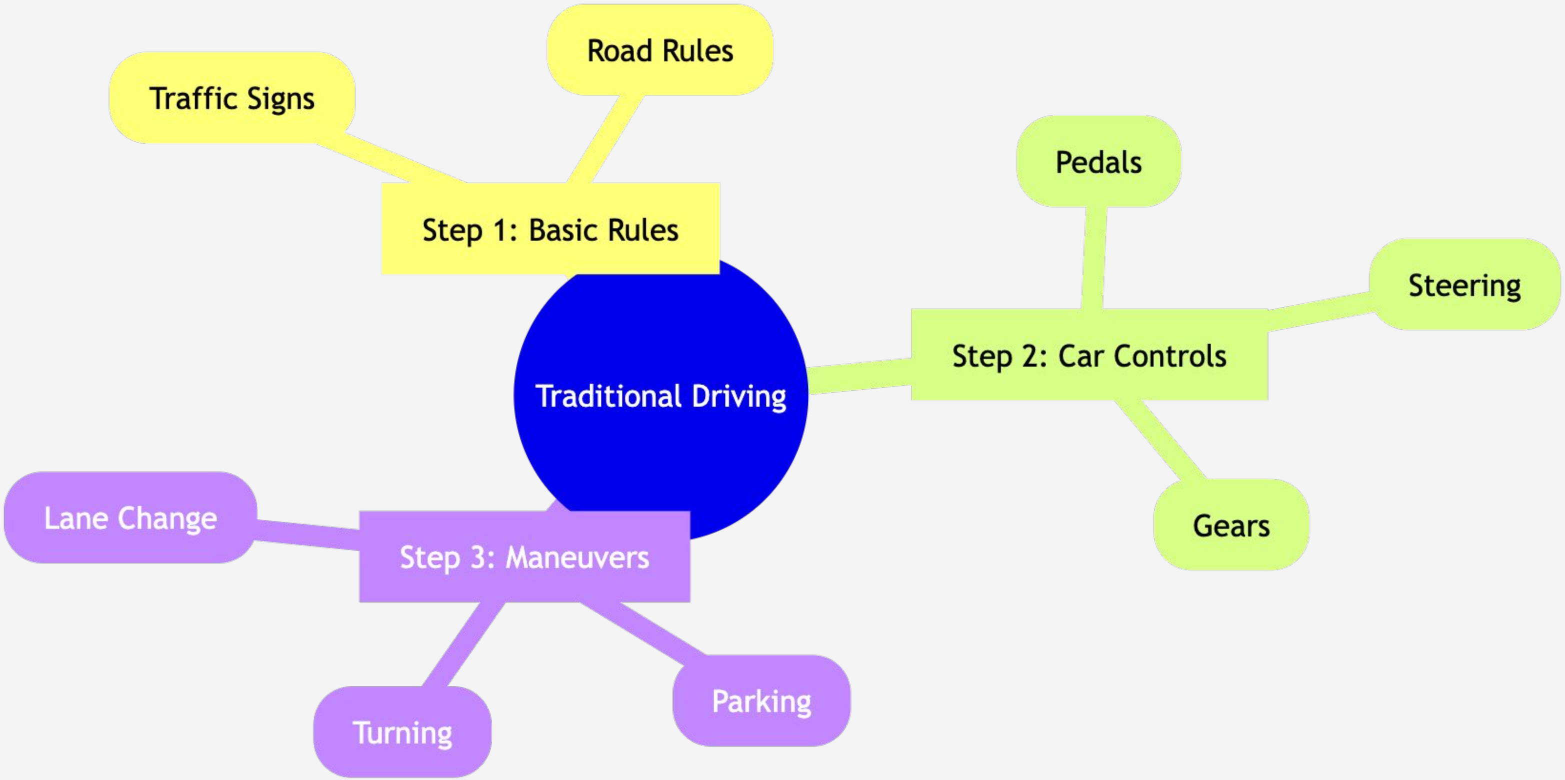
Deep Learning - Key aspects

3. End-to-End Learning - (analogy) Understanding End-to-End Learning 🚗



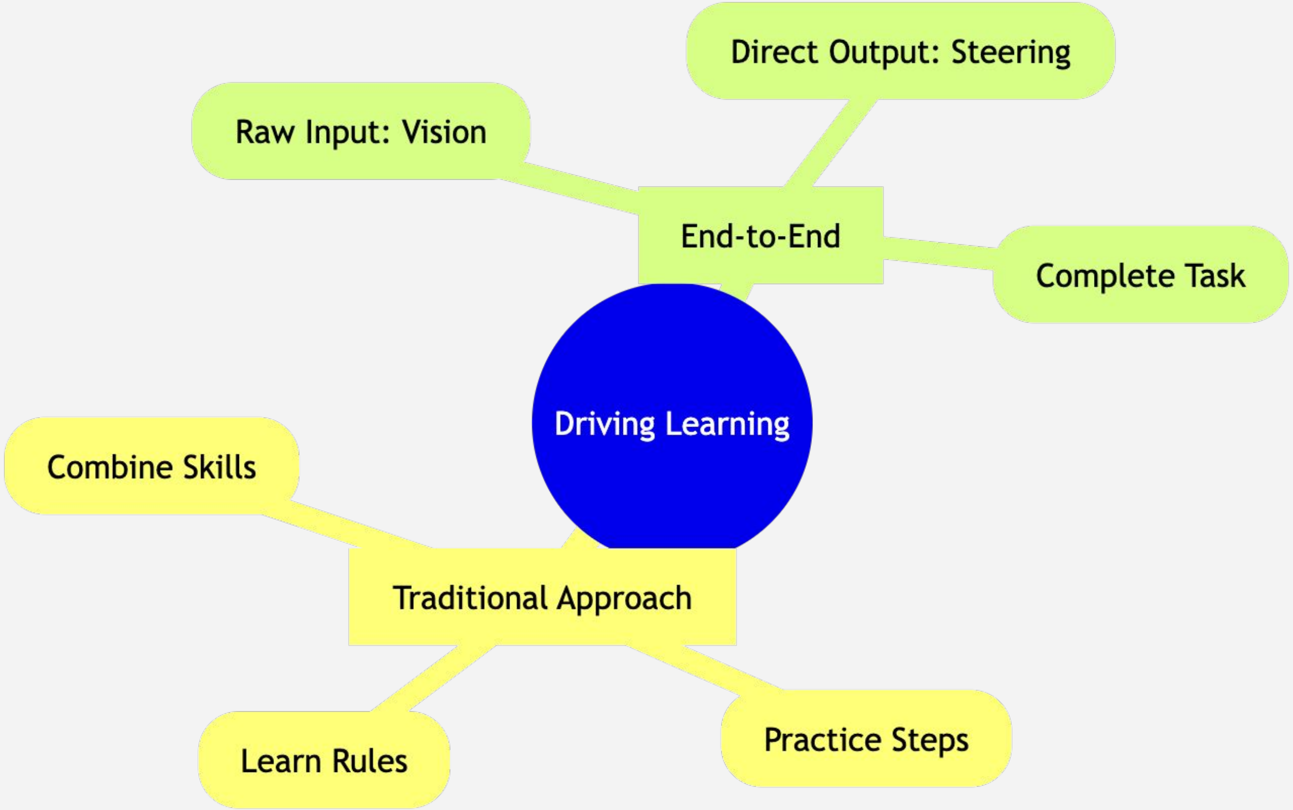
Deep Learning - Key aspects

Traditional driving - Breakdown



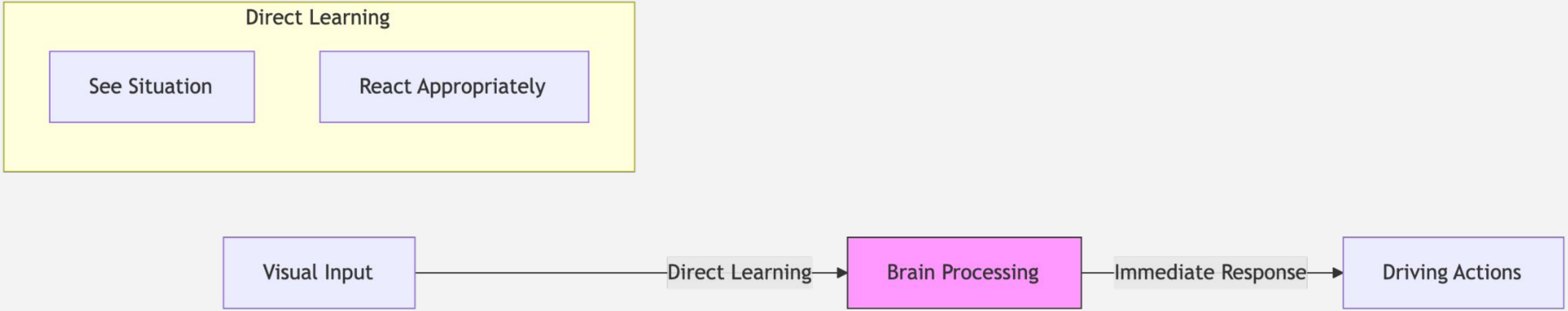
Deep Learning - Key aspects

3. End-to-End Learning - (analogy) Understanding End-to-End Learning 🚗



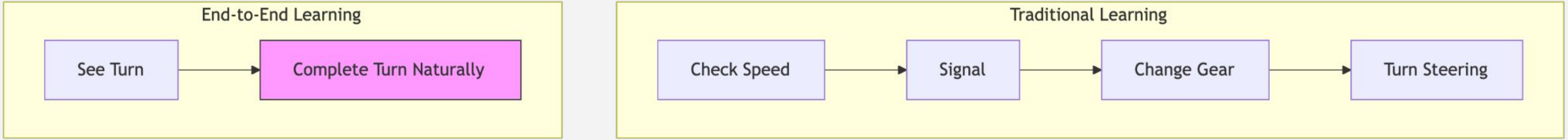
Deep Learning - Key aspects

3. End-to-End Learning - (analogy) Understanding End-to-End Learning 🚗

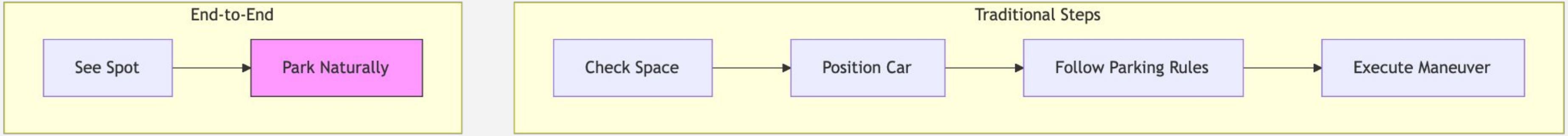


Comparison Through Scenario

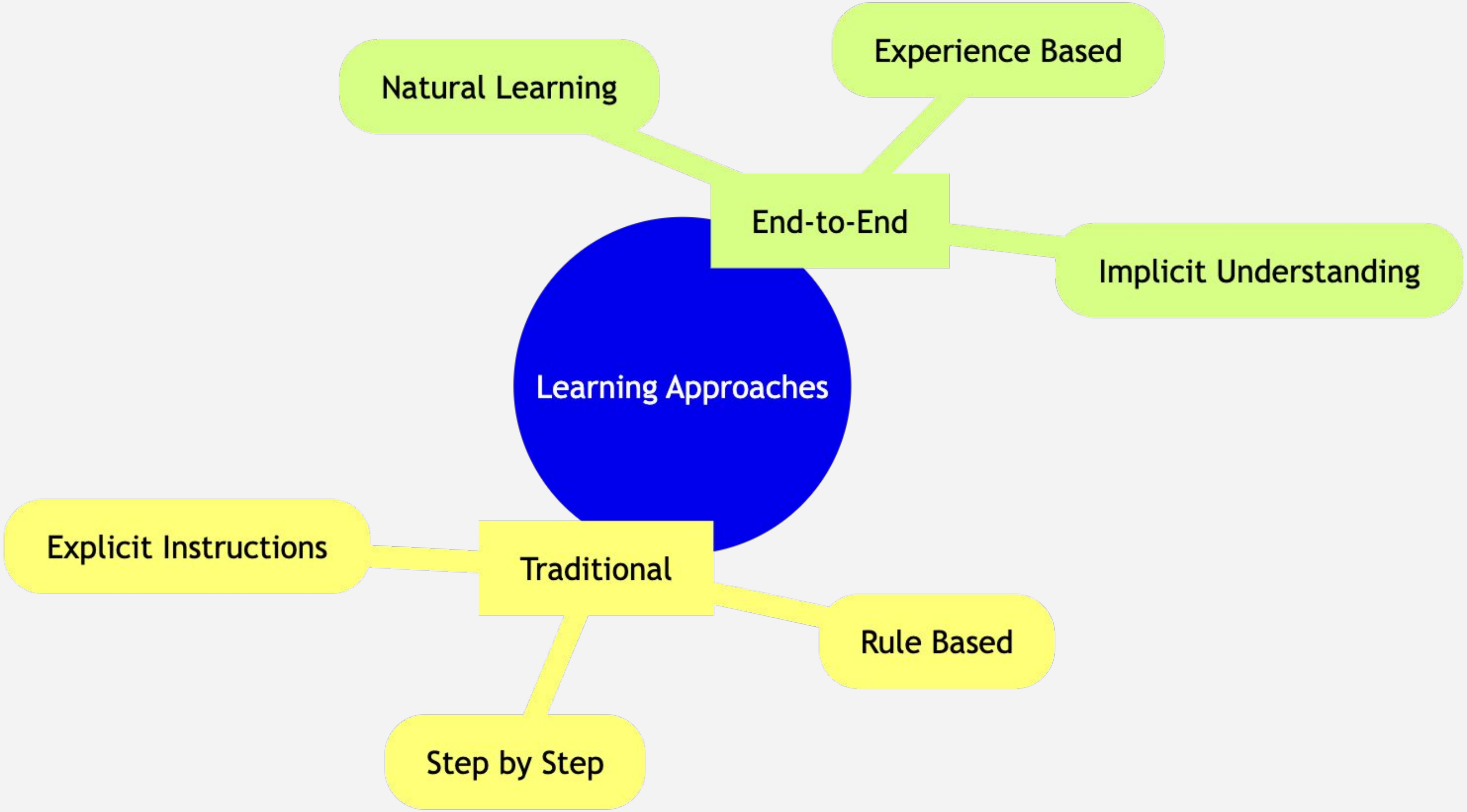
Scenario 1: Approaching a Turn



Scenario 2: Parking



Key Differences



Deep Learning - Real-World Analogy

Deep Learning Parallel

Pixels → Edges → Patterns → Objects → Scenes

Child Learning Process

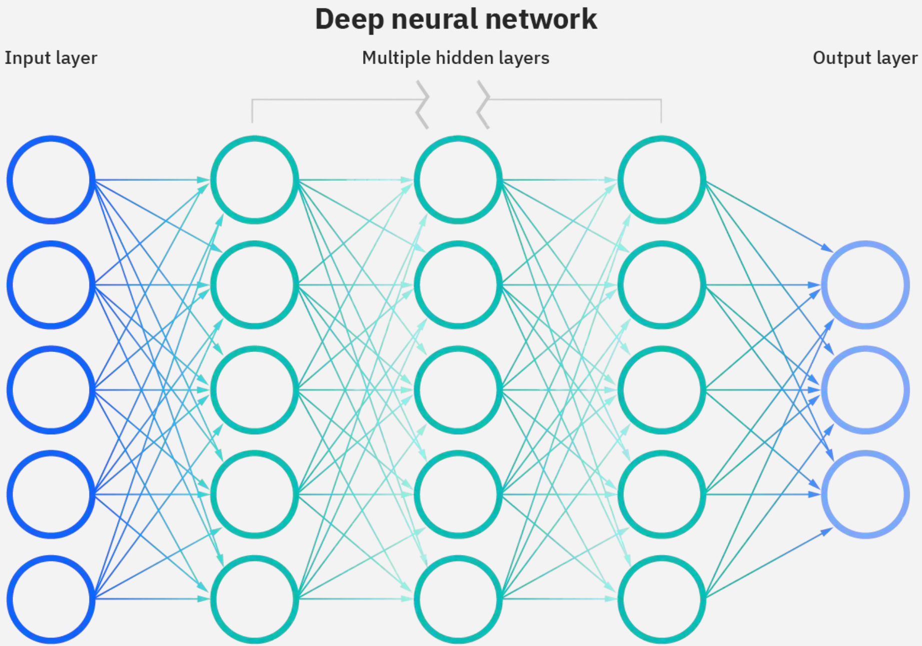
Basic Shapes → Letters → Words → Sentences → Stories

Deep Learning - Real-World Analogy

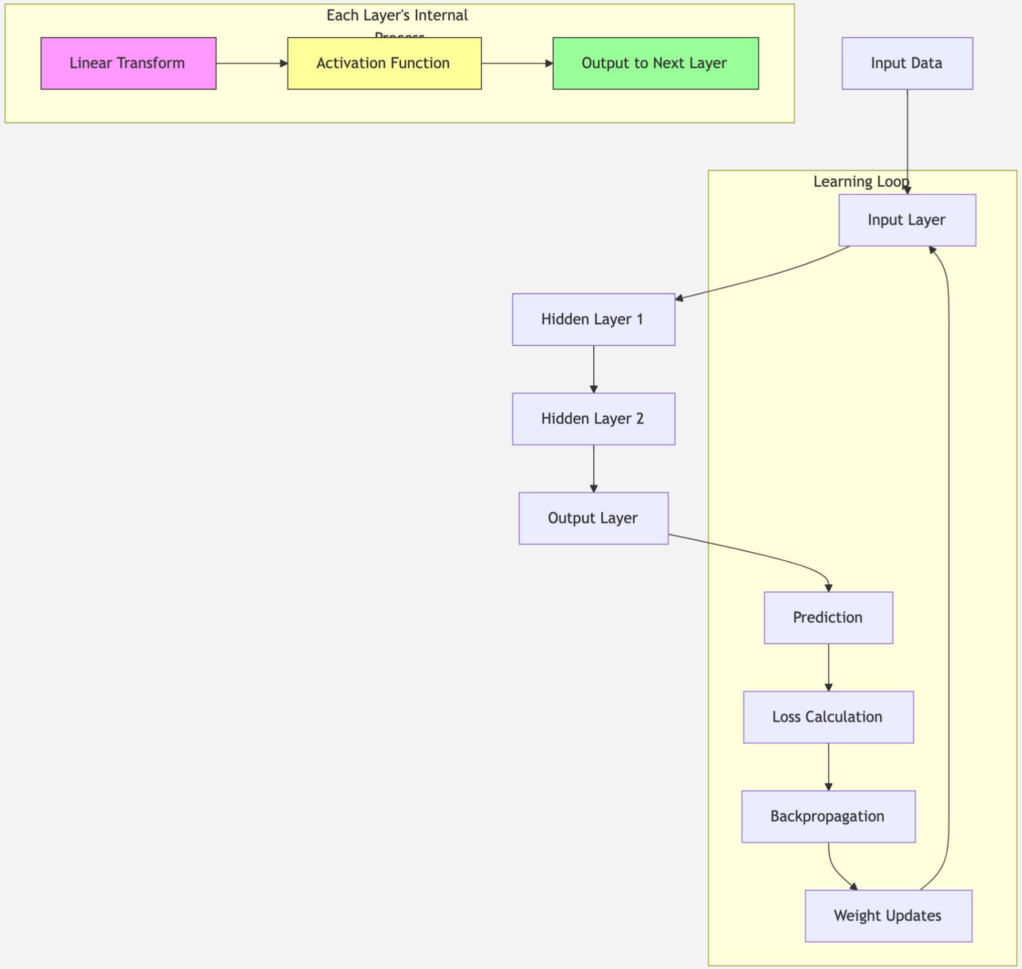
Deep Learning (DL)

Deep Learning

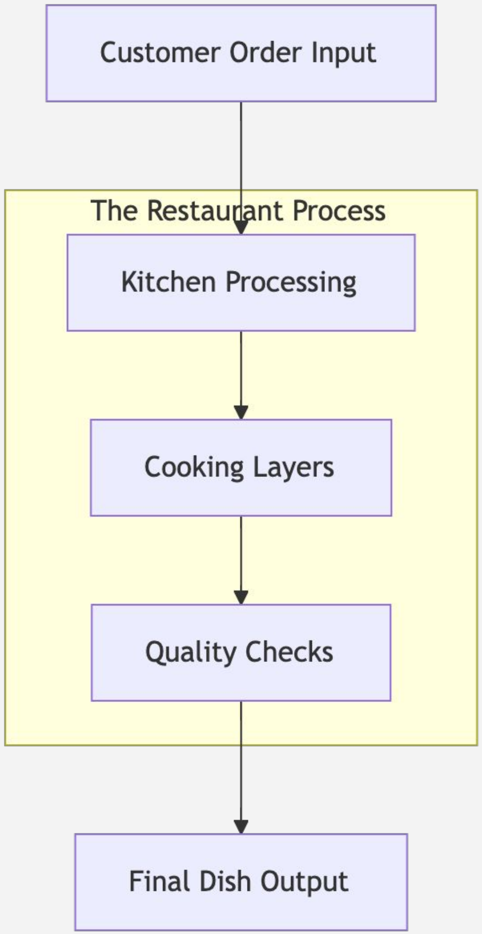
- Uses neural networks (computational models inspired by the human brain.



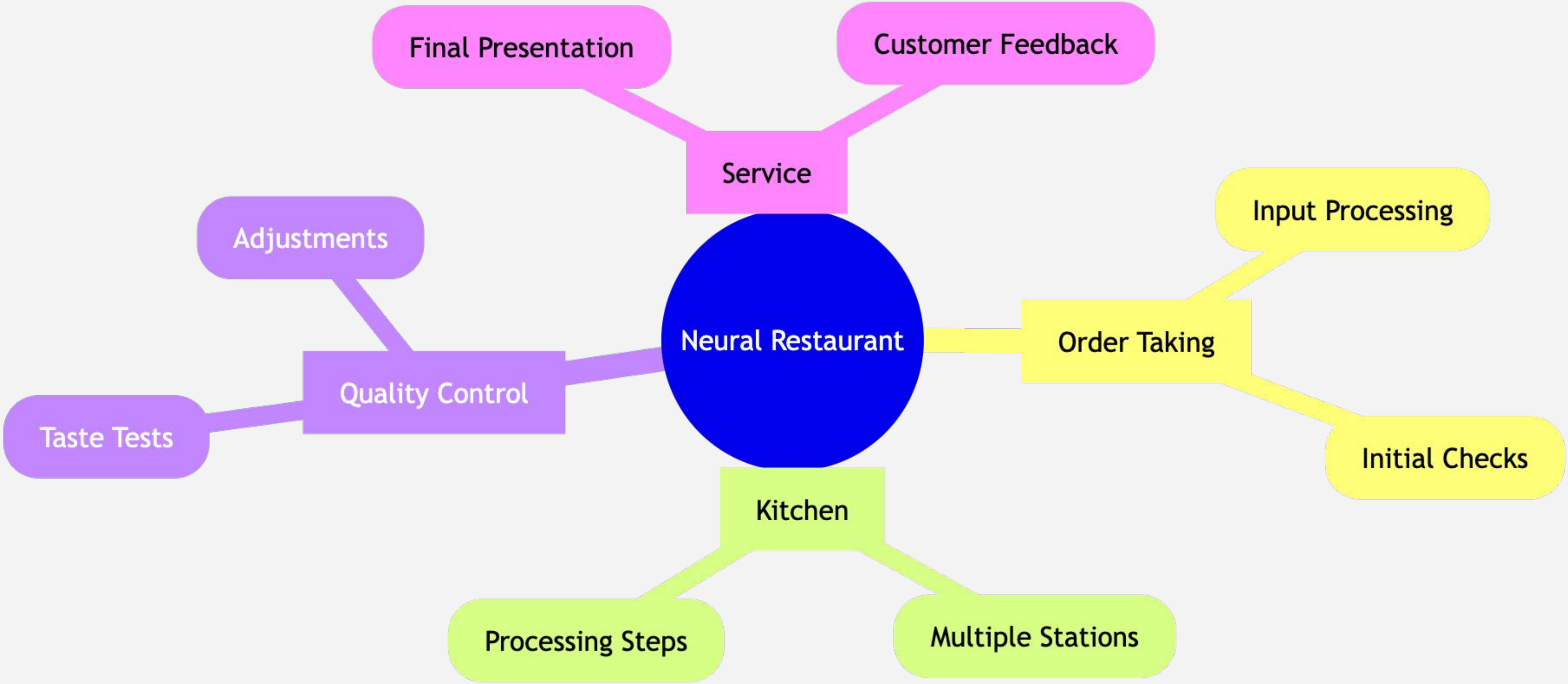
The Neural Network - Deep Dive



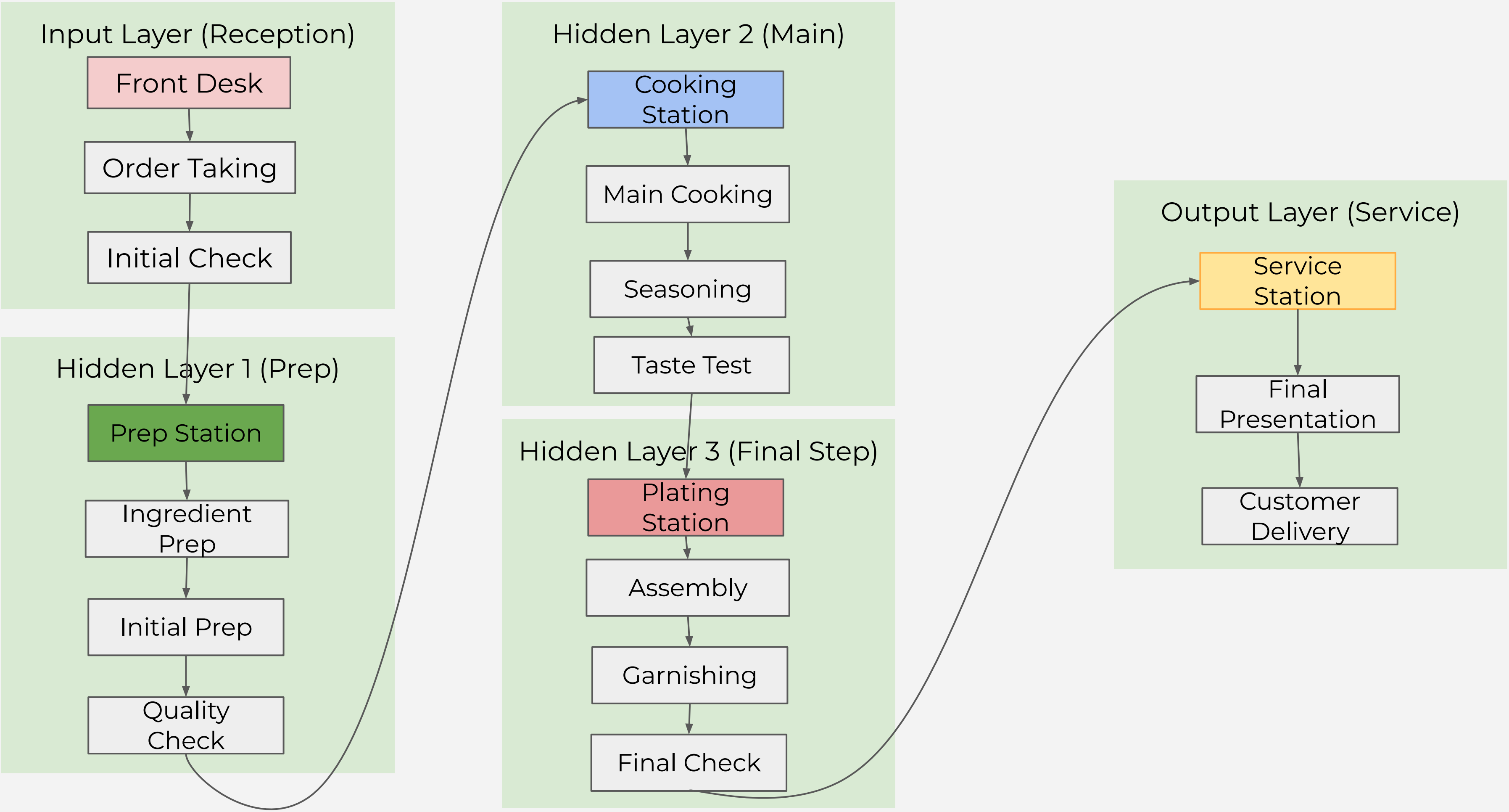
The Neural Network - Deep Dive



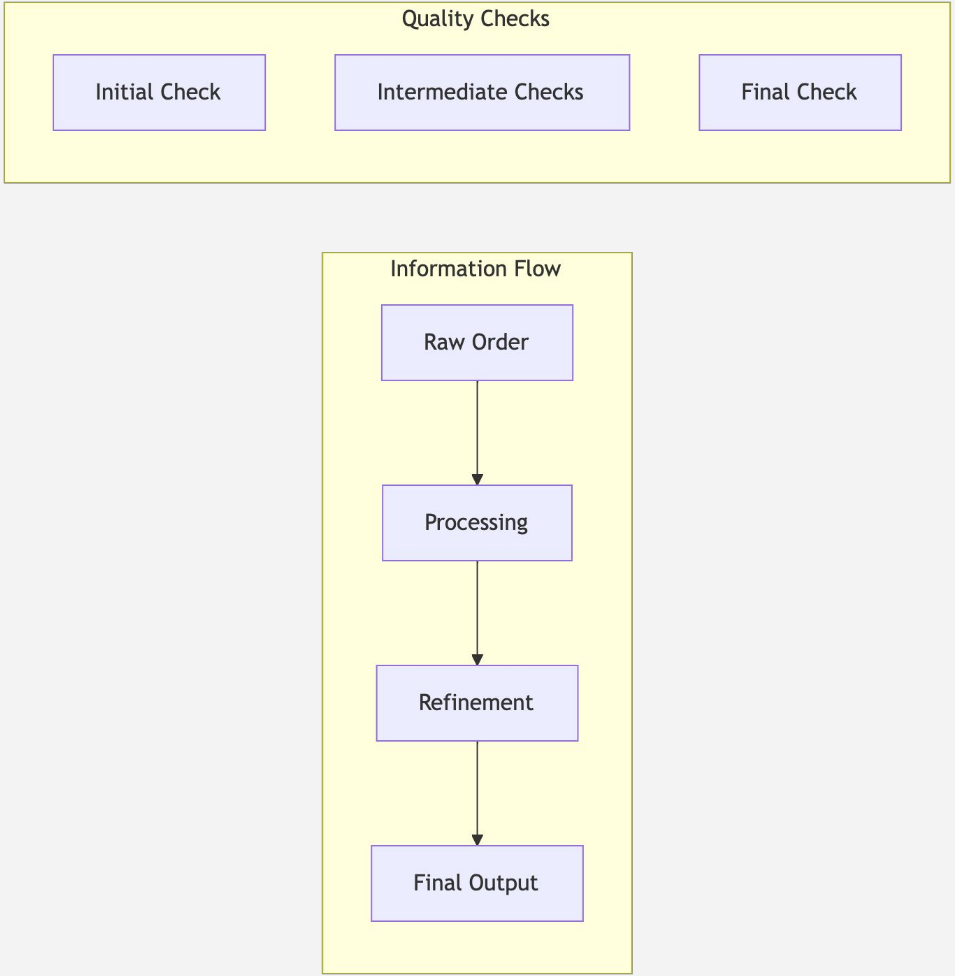
The Neural Network - Deep Dive



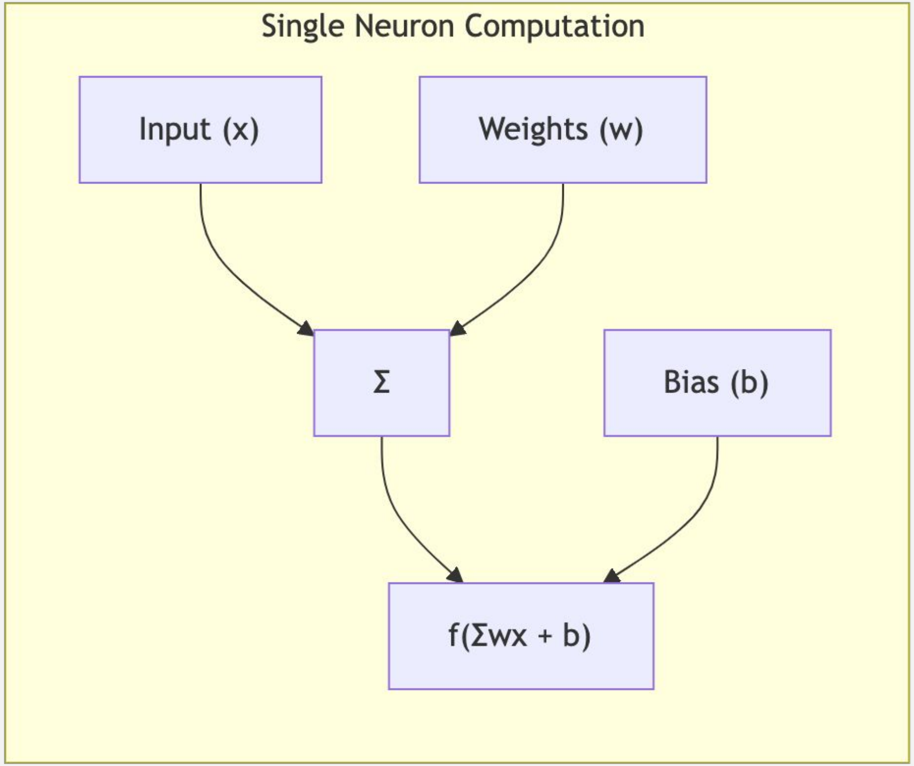
Network Structure (Restaurant Layout)



Network Structure (Restaurant Layout)

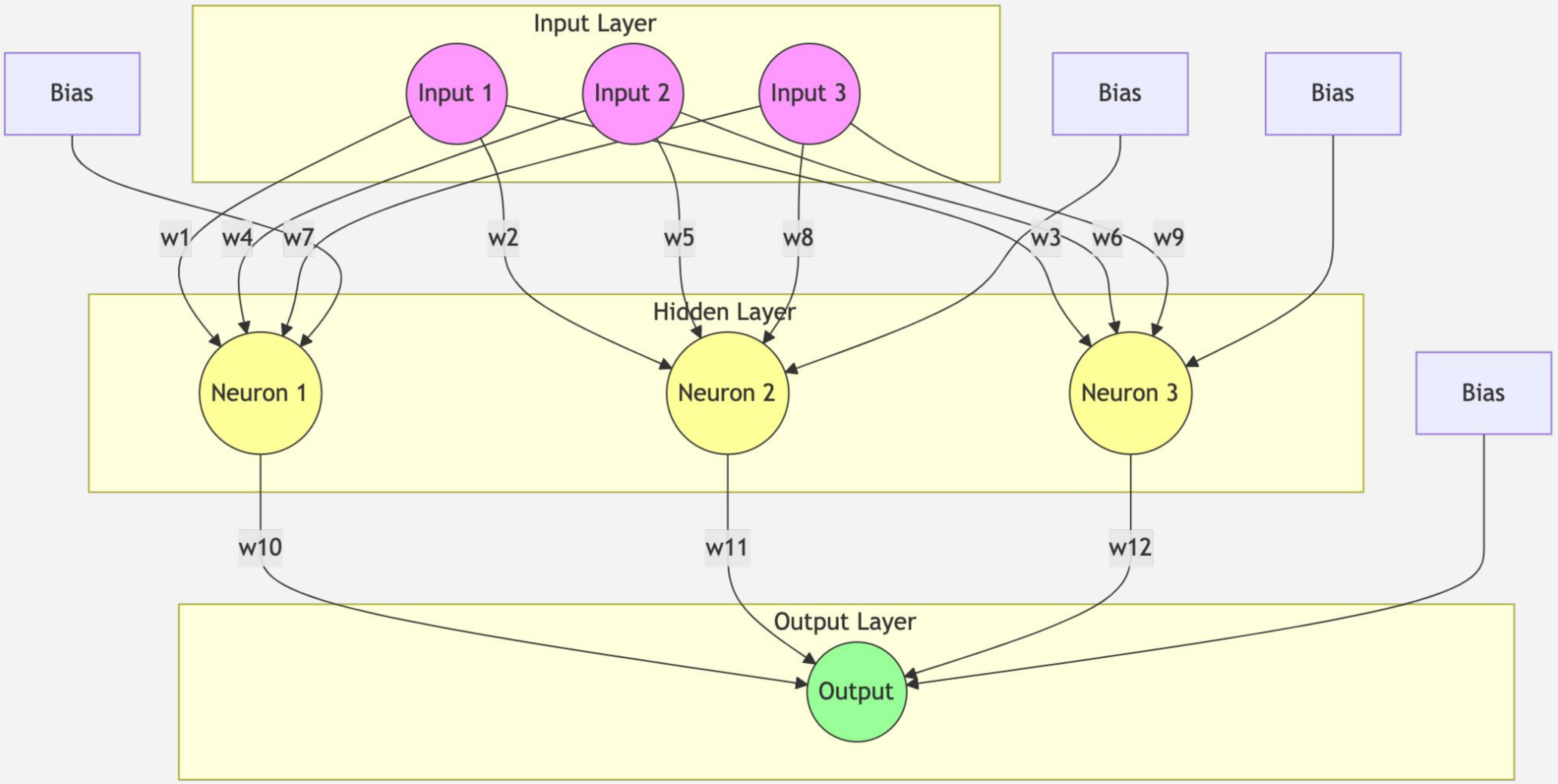


Neural Network Mathematical Foundations: Weights & Biases

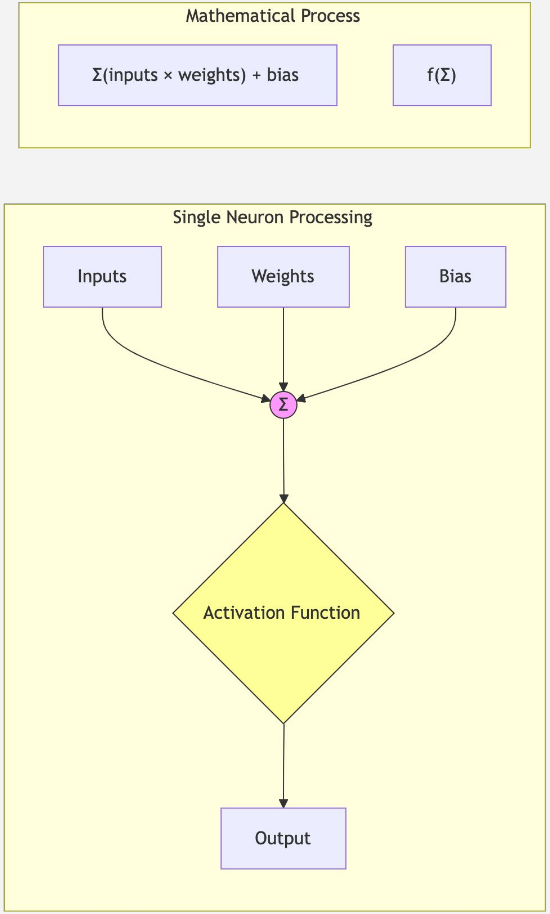


$$y = f\left(\sum_{i=1}^{n} w_i x_i + b\right)$$

Neural Network Key Concepts (Restaurant Analogy)

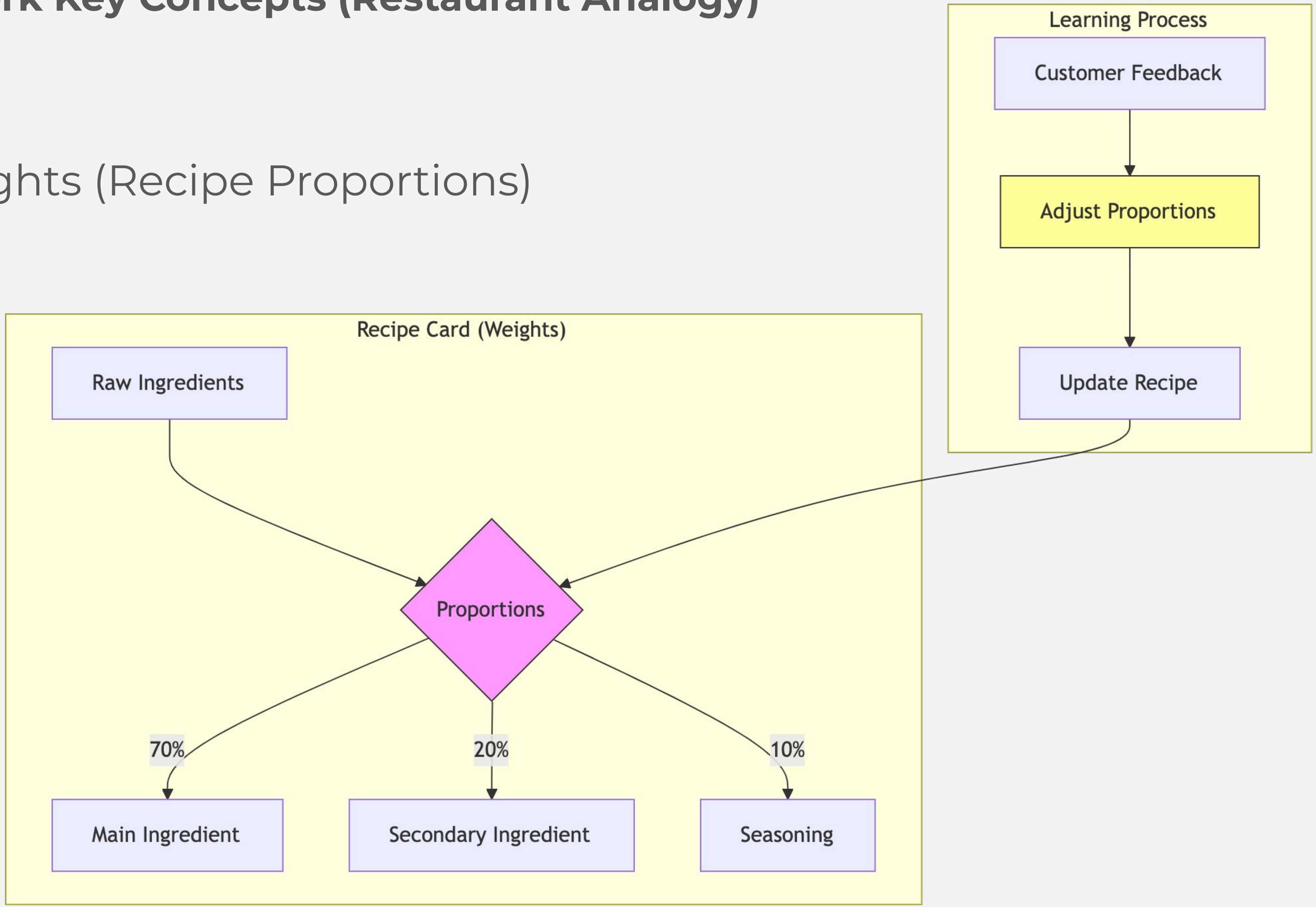


Neural Network Key Concepts (Restaurant Analogy)

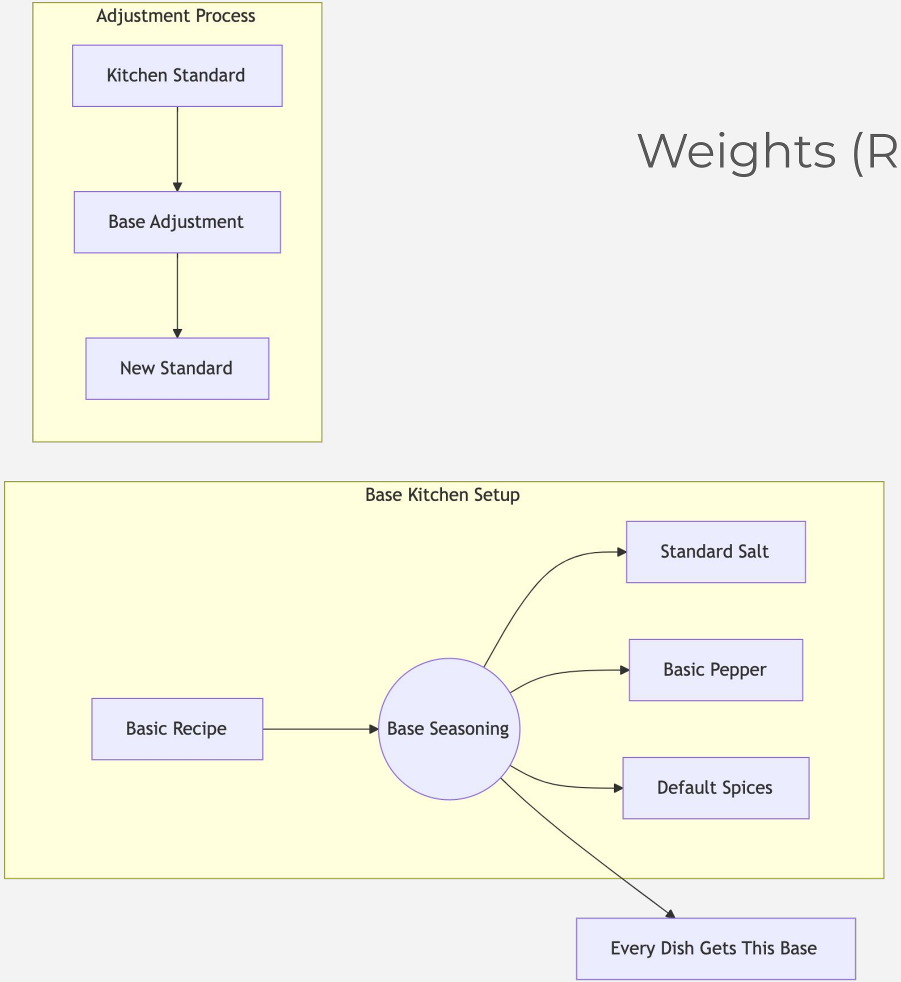


Neural Network Key Concepts (Restaurant Analogy)

Weights (Recipe Proportions)



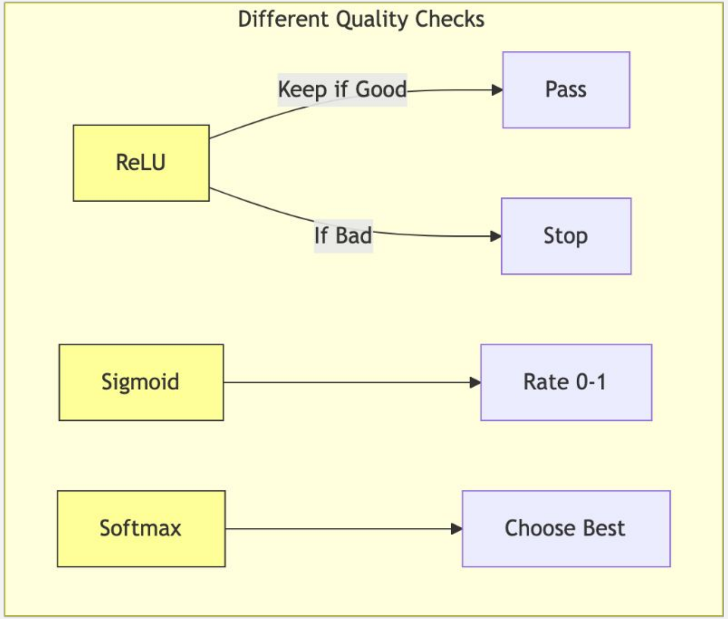
Neural Network Key Concepts (Restaurant Analogy)

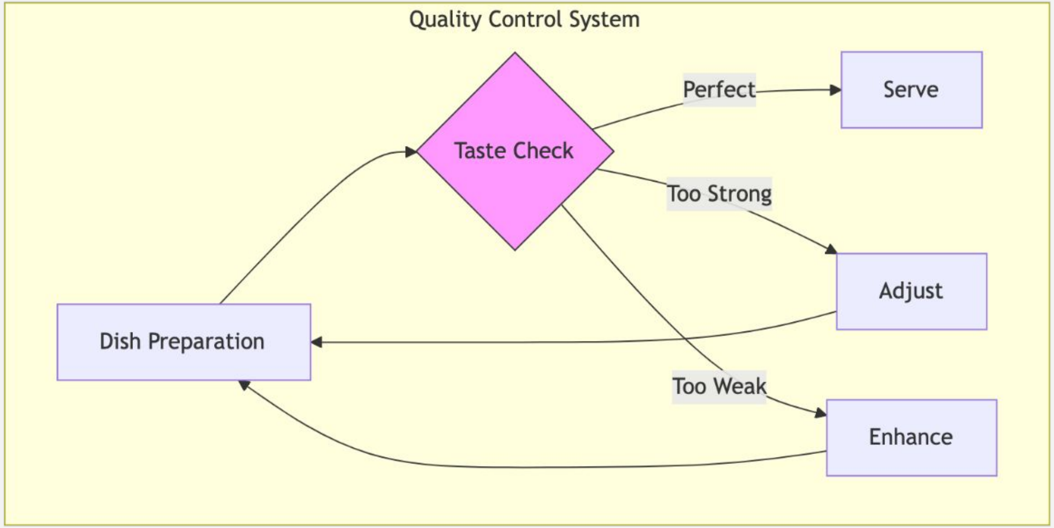


Weights (Recipe Proportions)

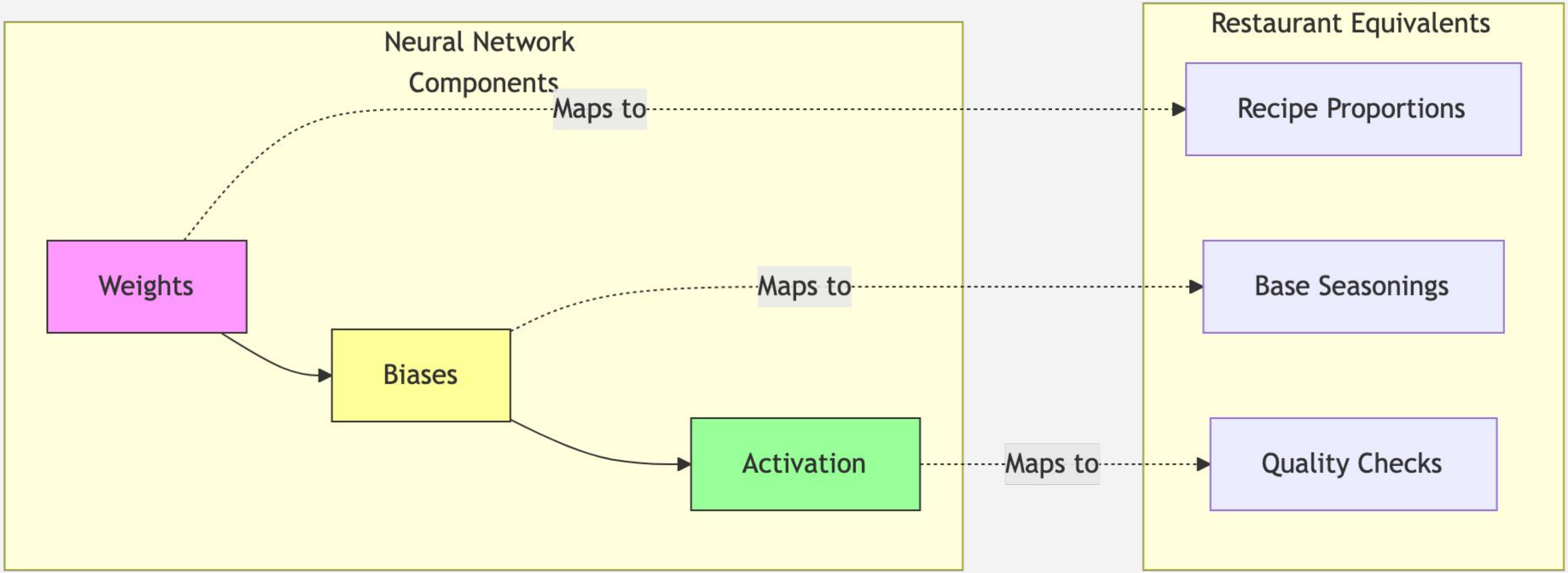
Neural Network Key Concepts (Restaurant Analogy)

Activation Functions (Quality Control)



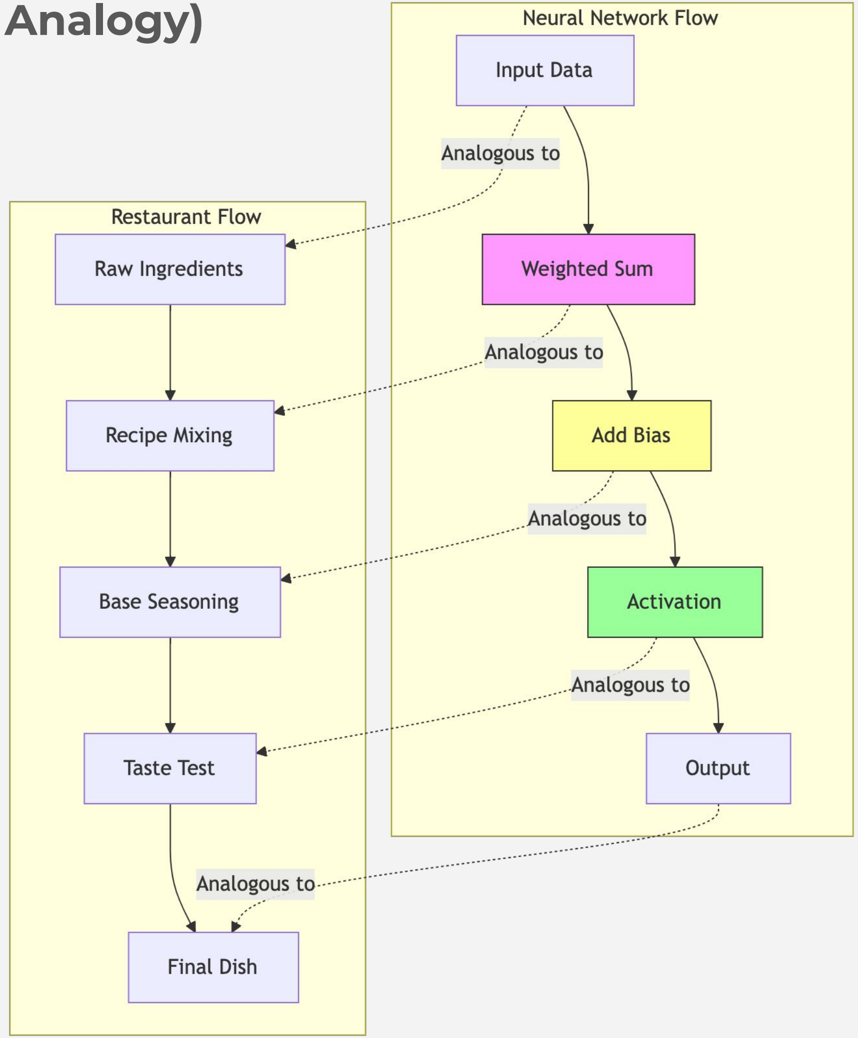


Neural Network Key Concepts (Restaurant Analogy)



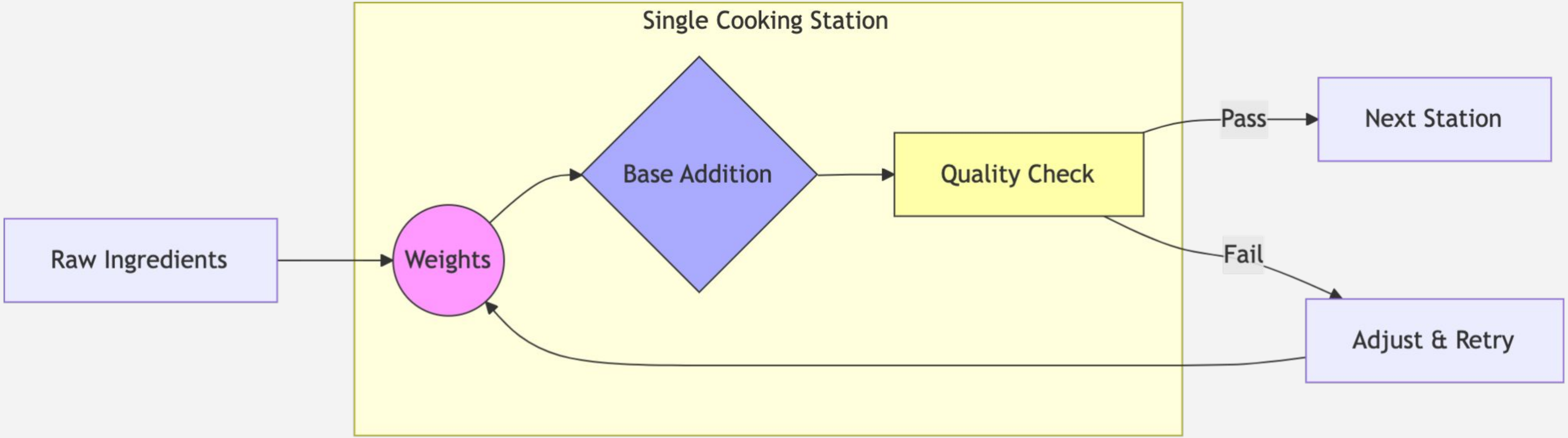
Neural Network Key Concepts (Restaurant Analogy)

Information Flow Comparison

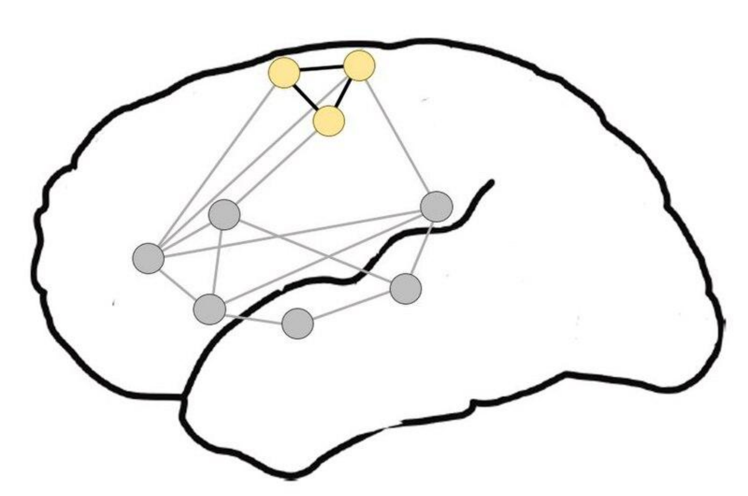


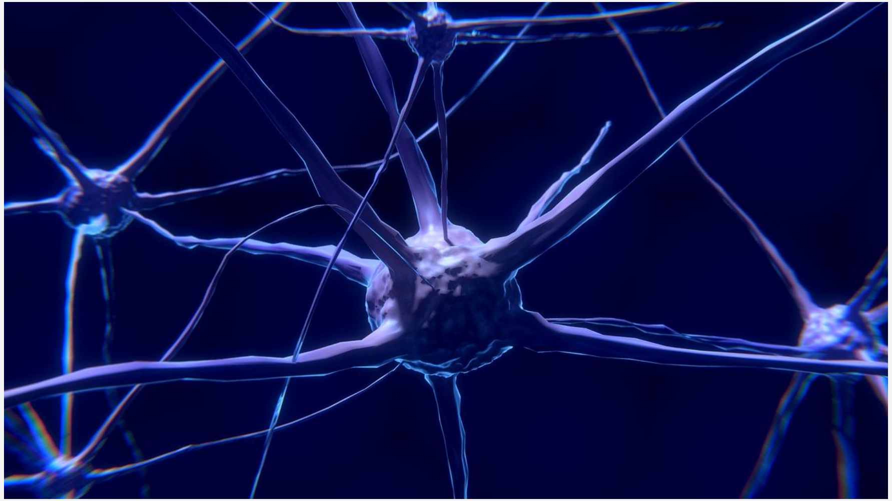
Neural Network Key Concepts (Restaurant Analogy)

Real-world example



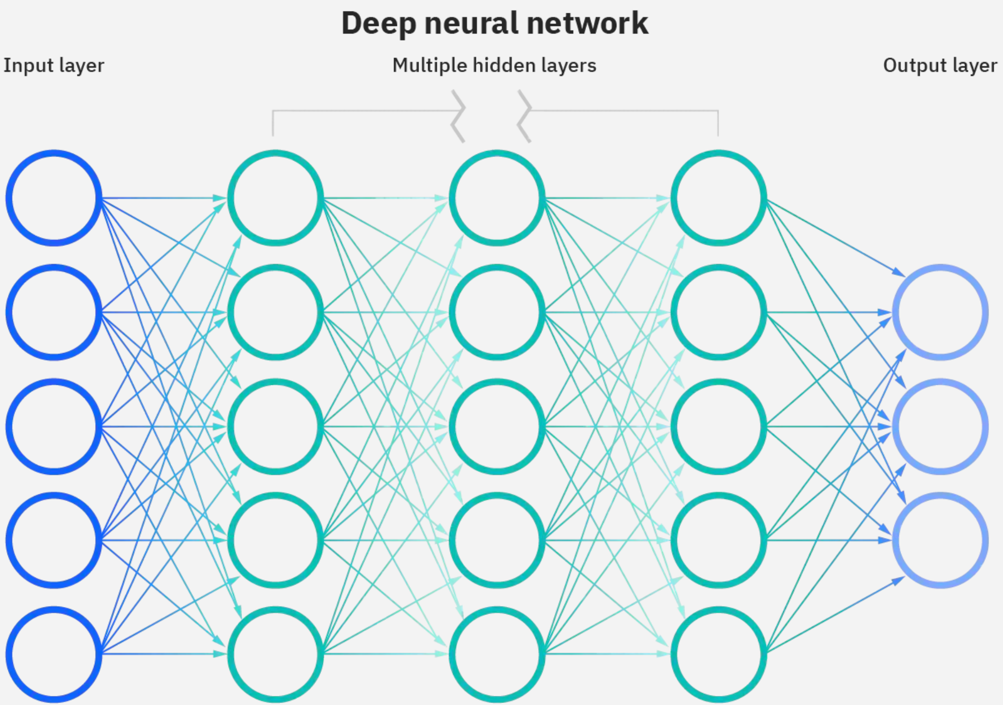
Neural Network - What is It?





Our brain has a network of neurons which are interconnected - they *process* information - make sense of the world around us.

Neural Network - AI



An AI Neural Network resembles the Brain Neural network!

Image source: https://zahere.com/how-chatgpt-works-the-architectural-details-you-need-to-know